

Deepak Sandrana

Data Engineer

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SUMMARY

- **Data Engineer** with 5+ years of experience designing data-intensive applications leveraging **Cloud Data Engineering, Data Warehousing, Hadoop Ecosystem, Big Data Analytics, Data Visualization, and Data Quality solutions.**
- Expert in developing interactive dashboards using **AWS QuickSight, Power BI, and Tableau**, visualizing key performance indicators (KPIs) from **Amazon Redshift** data for business stakeholders.
- Implementing a robust data streaming architecture using **Kafka**, efficiently handling the ingestion and distribution of real-time data across diverse applications.
- Skilled in RDBMS concepts, Data Modeling (Facts and Dimensions, Star/Snowflake schemas), Data Migration, and ETL Processes.
- Leveraged **Apache Spark** and **Spark SQL** within **Databricks** to develop data pipelines for customer usage analysis and extracted, transformed, and aggregated data from diverse file formats, uncovering valuable insights into customer behavior patterns.

EXPERIENCE

Capital One Financial, TX

Jan 2023 – Present

Data Engineer

- Led the development and optimization of **ETL pipelines** using **Python, SQL, Apache Spark, and Hadoop** to ensure high-fidelity migration of critical financial data with minimal disruption.
- Accelerated **Snowflake's** cloud-based architecture to facilitate collaboration between data analysts and data scientists, leading to a 30% increase in data-driven project completion rate.
- Built and deployed **Spark scripts** ETL (Extract, Transform, Load) jobs using **Glue crawlers** and, automating data ingestion from various sources into a data warehouse, saving 2 hours of manual workload per week.
- Integrated Apache **Airflow** with **AWS** to monitor multi-stage **ML workflows**, with tasks running on **Amazon SageMaker**, and contributed to CI/CD solutions using **Git** and **Jenkins** for setting up and configuring the big data architecture on the **AWS cloud platform**.
- Developed and maintained various **DBT models** and macros, focusing on incremental model design and modularization, resulting in a 40% increase in data pipeline efficiency and a 25% reduction in pipeline execution time.
- Leveraged **AWS Glue** to automate **ETL processes** for daily sales data feeds, reducing processing time by approximately 50%.

Fusion Software Technologies, India

Aug 2018 – Jul 2021

Data Engineer

- Built various **ETL/ELT** pipelines using **Databricks notebooks** and **SQL** to extract, transform, and load data from multiple sources into the data warehouse and lakehouse environments.
- Accomplished interactive **Power BI reports** leveraging advanced **DAX** and **Power Query** modeling techniques to unlock deeper data insights and facilitate informed decision-making.
- Orchestrated a complex data processing workflow consisting of multiple stages (data extraction, transformation, loading) using **AWS Pipeline**, improving data pipeline reliability and 40% reduction in data processing errors.
- Created and deployed **Apache Flink** applications for processing high-volume streaming data, achieving a 25% improvement in processing speed compared to traditional batch processing methods.
- Implemented data quality frameworks with validation checks, reducing data inconsistencies by 35% and ensuring high accuracy for analytics and reporting.
- Crafted and modified complex SQL queries, leveraging indexing and query optimization techniques to enhance database efficiency, leading to a 30% improvement in application response time.
- Improved **ETL workflows** by rigorous testing and debugging of **SQL scripts** and Python code resulting in a 50% increase in data processing efficiency and ensuring seamless data integration with downstream systems.
- Streamlined **Hive queries** for performance optimization by activating techniques such as partitioning, bucketing, and compression to handle large datasets effectively, implemented data processing systems by **MapReduce** to handle workloads efficiently and accurately.

SKILLS

Programming Language: Python, R, SQL, Scala

Big Data Ecosystem: Hadoop, HDFS, MapReduce, HBase, Sqoop, Apache Kafka, Apache Spark, Flink, Zookeeper

Machine Learning: Linear, Logistic Regression, Decision Trees, Random Forests, K-Means

Cloud Technologies: AWS (EC2, S3, RDS, Lambda, Glue, Athena, AWS Pipeline, Redshift, DynamoDB)

ETL & Tools: SSIS, SSRS, Data Pipelines, Data build tool (DBT), Apache Airflow, Tableau, Power BI, Excel, Terraform

Packages & Data Processing: NumPy, Pandas, Matplotlib, Seaborn, SciKit-Learn, PySpark, Data Pipelines, Jenkins

Version Control & Database: GitHub, Git, SQL Server, PostgreSQL, MongoDB, MySQL, Snowflake, HiveQL

EDUCATION

Master of Science in Computer Science

University of Texas at Arlington, Texas

Bachelor in Computer Science

Indian Institute of Information Technology, Vadodara, India